Reference Sheet for Exam 2

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Hi this is Johnny! The person who wrote this document, due to the restriction placed on by the professor we are not allowed to use the same personal equation sheet, I am more than happy for you to copy my cheat sheet so you can make your own, you can even copy the formating if you like. Sorry for the inconvenience.

This is stil a work in progress I am finishing some stuff up based on suggestions of other friends.

13. Electric Field

Electric fields can be thought as a shared universal plane that is influenced by charges and influences charges, similar to gravity and mass.

13.1 Coulomb's Force Law for Point Particles

Describes the force behaviour that two charges experience.

$$|\overrightarrow{F}| = F = \frac{1}{4\pi\epsilon_0}\frac{|Q_1Q_2|}{r^2}$$

13.3 Definition of Electric Field

This equation says that the force on particle 2 is determined by the charge of the particle 2 and by the electric field $\overrightarrow{E}_1$ made by all other charged particles.

$$\overrightarrow{F}_2 = q_2\overrightarrow{E}_1$$

13.4 Electric Field of a Point Particle

This equation represents how the electric field is influenced due to a charge, with the use of calculus, it can be used to find the contributions of multiple electrical fields at a determinent point.

$$\overrightarrow{E}_1 = \frac{1}{4\pi\epsilon_0}\frac{q_1}{|\overrightarrow{r}|^2}\hat{r}$$

Continuous Charge Distributions

If we have objects in higher dimensions, the charge is distrbuted along them continously.

- Linear Density (1D, for lines and curves), $\lambda = \frac{dQ}{dl}$ in (C/m).
- Surface Density (2D, for planes and surface), $\sigma = \frac{dQ}{dA}$ in (C/m^2)
- Volume Density (3D, for solids and volumes), $\rho = \frac{dQ}{dV}$ in (C/m^3).

Similarly to the total charge for a point is q, for the total charge of higher dimensional objects is the integral of the charge density times the dimensional measure.

$$Q = \int \lambda dl \quad Q = \int \sigma dA \quad Q = \int \rho dV$$

Electric Field from Continuous Distributions

Starting from a point-charge field replace dQ with the appropiate dimensional density.

$$d\overrightarrow{E} = \frac{1}{4\pi\epsilon_0}\frac{dQ}{r^2}\hat{r} \rightarrow \overrightarrow{E} = \frac{1}{4\pi\epsilon_0}\int\frac{dQ}{r^2}\hat{r}$$

Infinite Line of Charge (uniform λ)

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Ring of a Charge (uniform λ)

$$E = \frac{1}{4\pi\epsilon_0}\frac{Qr}{(r^2+R^2)^{\frac{3}{2}}}\hat{r}$$

Note that R is the radious of the ring, r is the distance along the axis (perpendicular to the plane of the ring) from the center